

Helix Thready — Component Reusability Platform Map

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Rev	Date	Author	Change
1	2026-07-22	swarm (design)	Initial matrix: every library component × 8 platform realizations; <code>.ds-*</code> names re-verified at source via <code>gh</code> (design_system, UI-Components-React, UI-Components-KMP, helix_design trees fetched 2026-07-22); per-component customization notes; VERIFIED/ ASSUMED discipline
2	2026-07-22	swarm (design · library)	Second verification pass (<code>gh</code> , file contents): React <code>Button.tsx</code> styling confirmed not token-bound (hard-coded Tailwind palette); KMP <code>Theme.kt</code> confirmed branded for another product (Yole, Material-red) — both recorded in §2; platform fan-out diagram added (§2.1 + <code>diagrams/platform-fanout.mmd</code>)

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1. Scope & verification method

This map records, for every component in the living library ([components.html](#)), how it is realized on each target platform of the decision matrix: **Angular** (`.ds-*` from `vasic-digital/design_system`), **React** (`UI-Components-React`), **Compose** (`UI-Components-KMP`), **Flutter** (`helix_design`), **SwiftUI**, **ArkTS** (HarmonyOS), **Qt** (Aurora), and **TUI** (Bubble Tea + Lipgloss).

Markers (per CONVENTIONS §7 — no bluff):

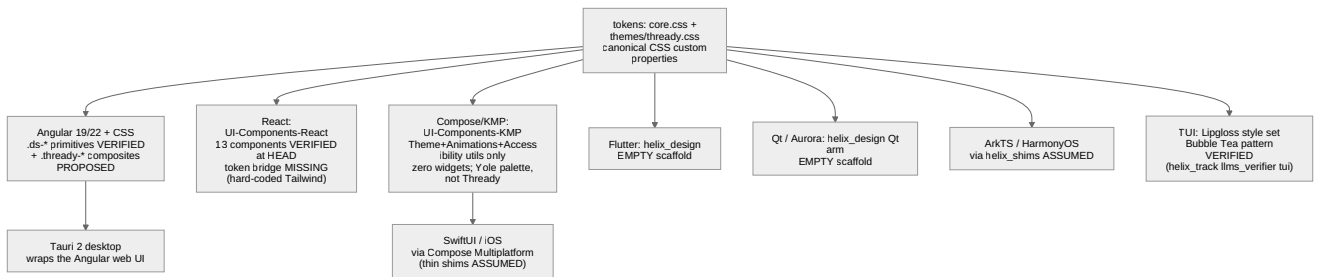
- **VERIFIED** — the named class/file exists; checked on **2026-07-22** by fetching the repo tree/content via `gh api` (`vasic-digital/design_system` , `vasic-digital/UI-Components-React` , `vasic-digital/UI-Components-KMP` , `vasic-digital/helix_design`) or by reading the local clone (`helix_track/llms_verifier/llm-verifier/tui/*.go` imports `charmbracelet/lipgloss`).
- **PROPOSED** — a `[DEFAULT — adjustable]` name/realization this design area proposes (upstream-contribution candidate); the CSS exists **only** in `components.html` , not upstream.
- **ASSUMED** — the platform package (or the widget inside it) does not exist or could not be verified; the cell records the *intended* native mapping. Nothing in an ASSUMED cell may be claimed to work `[GAP: 8.2/8.3/8.4/8.5/8.6]` .

2. Per-repo verification results

Repo / source	What was verified (2026-07-22)	Status
<code>vasic-digital/design_system</code> → <code>components/css/components.css</code>	Complete <code>.ds-*</code> set at HEAD (106 lines): <code>.ds-container</code> , <code>.ds-section</code> , <code>.ds-btn</code> (+ <code>--primary/--secondary/--ghost</code> , <code>:hover/:active/:focus-visible</code> , <code>reduced-motion</code>), <code>.ds-card</code> (+ <code>--raised</code>), <code>.ds-input</code> , <code>.ds-link</code> , <code>.ds-nav</code> (+ <code>__links/__link</code> , <code>aria-current</code>), <code>.ds-footer</code> , <code>.ds-badge</code> (+ <code>--success/--warn/--danger</code>), <code>.ds-brand-mark</code>	VERIFIED — production-usable web foundation [GAP: 8.1]
<code>vasic-digital/design_system</code> → <code>components/angular/*</code>	<code>theme.service.ts</code> , <code>il8n.service.ts</code> , <code>theme-toggle.component.ts</code> , <code>language-picker.component.ts</code> , <code>ds.config.ts</code> present at HEAD	VERIFIED
<code>vasic-digital/UI-Components-React</code> → <code>src/components/*</code>	Component files at HEAD: <code>Avatar</code> , <code>Badge</code> , <code>Button</code> , <code>Card</code> , <code>EmptyState</code> , <code>ErrorBoundary</code> , <code>Input</code> , <code>LoadingSpinner</code> , <code>Progress</code> , <code>Select</code> , <code>Switch</code> , <code>Tabs</code> , <code>Textarea</code> (<code>.tsx</code> , each with a test). <code>Button.tsx</code> content inspected: variants <code>primary\ secondary\ outline\ ghost\ destructive</code> , sizes <code>sm\ md\ lg</code> , a <code>loading</code> spinner prop — but <code>variantClasses</code> hard-codes a Tailwind palette (<code>bg-blue-600</code> , gray neutrals, <code>focus-visible:ring-blue-500</code>); no design-system token is referenced	VERIFIED files; VERIFIED not token-bound — the package is <code>SCAFFOLD/FLAGGED</code> [GAP: 8.6] : the blocking work is the token bridge (re-bind variants to <code>tailwind-v4.css</code> / <code>var(--accent)</code> , map <code>destructive</code> → <code>--danger</code>), then re-audit
<code>vasic-digital/UI-Components-KMP</code> → <code>src/commonMain/kotlin/digital/vasic/uicomponents/*</code>	Only <code>Theme.kt</code> , <code>Animations.kt</code> , <code>Accessibility.kt</code> (+ tests). No widget components exist. <code>Theme.kt</code> content inspected: <code>object YoleColors</code> with <code>BrandPrimary = Color(0xFFD32F2F)</code> (Material Red 700) — the palette belongs to another product (“Yole”) , not Thready, and references no design-system token	VERIFIED as utilities-only scaffold, foreign-branded [GAP: 8.4] — every widget cell below is ASSUMED; a <code>ThreadyColors</code> token-bridge codegen must replace the hand-kept palette before any use

Repo / source	What was verified (2026-07-22)	Status
<code>vasic-digital/helix_design</code>	Tree at HEAD contains only <code>README.md</code> , agent config (<code>CLAUDE.md</code> , <code>AGENTS.md</code> , ...), <code>upstreams/*.sh</code> . No Flutter/Qt packages at all.	VERIFIED as empty scaffold [GAP: 8.2/8.3] — every Flutter/Qt cell is ASSUMED
TUI pattern (local clone)	<code>helix_track/llms_verifier/llm-verifier/tui/{app.go,notifications.go}</code> import <code>charmbracelet/lipgloss</code> ; <code>screens/</code> + <code>cmd/tui</code> exist	VERIFIED pattern — Thready styles themselves are PROPOSED (generated from the token export, design-system.md §7)
SwiftUI package	No in-house SwiftUI UI package is named in the decision matrix or found under <code>vasic-digital</code>	ASSUMED entire column [OPEN: THREADY-DES-LIB-02]
ArkTS / Qt (Aurora)	Only reachable via native clients + <code>helix_shims</code> per ground truth; <code>helix_shims</code> interface not inspected	ASSUMED entire columns [GAP: 8.5]

2.1 Platform fan-out (diagram)



Rendered PNG/SVG exported via Docs Chain (\$11.4.65). Source: `diagrams/platform-fanout.mmd` (rendered sibling: `diagrams/platform-fanout.svg`).

Explanation (for readers/models that cannot see the diagram). A single node sits at the top: the canonical token source, `tokens/core.css` plus `tokens/themes/thready.css` — the CSS custom properties specified in `design-system.md` §3. Eight arrows fan out of it, one per platform column of the matrix in §3. The **Angular/CSS** arrow is the production-usable path: the verified `.ds-*` primitives plus the proposed `.thready-*` composites; the **Tauri 2 desktop** node hangs off it because the desktop client wraps the same Angular web UI and needs no separate component work. The **React** arrow points at `UI-Components-React`, which really ships thirteen components but styles them with a hard-coded Tailwind palette — its missing piece is the token bridge, not the components. The **Compose/KMP** arrow points at `UI-Components-KMP`, which ships three utility files (theme, animations, accessibility) branded for another

product and no widgets; the **SwiftUI** node hangs off KMP because the sanctioned iOS path is Compose Multiplatform (or thin SwiftUI shims), not a hand-built SwiftUI kit. The **Flutter** and **Qt/Aurora** arrows point at `helix_design`, an empty scaffold — both are plans, not code. The **ArkTS** arrow is the HarmonyOS path through `helix_shims`, also a plan. The **TUI** arrow maps the token palette onto a Lipgloss style set following the verified in-house Bubble Tea pattern. Because every arrow starts at the same token file, a theme or white-label swap re-tints every platform without touching structure.

3. The matrix

Column legend — **NG**: Angular (Web/Desktop-Tauri, primary `[OPERATOR]`); **RE**: React (`UI-Components-React`); **CM**: Compose Multiplatform (`UI-Components-KMP`); **FL**: Flutter (`helix_design`); **SW**: SwiftUI; **AR**: ArkTS; **QT**: Qt/QML (Aurora); **TU**: TUI (Bubble Tea + Lipgloss). Cell = realization + marker (`VERIFIED` · `o` `PROPOSED` · `Δ` `ASSUMED`).

Component	NG Angular .ds-*	RE React	CM Compose	FL Flutter	SW SwiftUI	AR ArkTS	QT Qt/ QML	TU TUI
Container / section	<code>.ds-container</code> , <code>.ds-section</code>	layout div + tokens Δ	<code>Column</code> + <code>ThreadySpacing</code> Δ	<code>Padding</code> / <code>Container</code> Δ	<code>VStack</code> + spacing Δ	<code>Column</code> Δ	<code>ColumnLayout</code> Δ	Lipgloss <code>Width / Margin</code> Δ
Button primary/secondary/ghost	<code>.ds-btn</code> + <code>--primary</code> / <code>--secondary</code> / <code>--ghost</code>	<code>Button.tsx</code> (file) <code>[GAP: 8.6]</code>	<code>Button / OutlinedButton / TextButton</code> Δ	<code>FilledButton / OutlineButton / TextButton</code> Δ	<code>Button</code> + <code>.buttonStyle</code> Δ	<code>Button</code> Δ	<code>Button</code> + QML style Δ	Lipgloss bold accent-bg style Δ (pattern)
Button destructive	<code>.ds-btn--danger</code> Δ	<code>Button</code> variant prop Δ	<code>Button(colors=error)</code> Δ	<code>FilledButton.errorColors</code> Δ	<code>Button(role: .destructive)</code> Δ	<code>Button.warnStyle</code> Δ	<code>Button.dangerPalette</code> Δ	danger-fg style Δ
Button loading/disabled	<code>[disabled]</code> + <code>.ds-btn--loading</code> Δ	<code>Button</code> + <code>LoadingSpinner.tsx</code> (files)	<code>enabled=false</code> + <code>CircularProgressIndicator</code> Δ	<code>onPressed: null</code> + spinner Δ	<code>.disabled()</code> + <code>ProgressView</code> Δ	<code>enabled</code> + <code>LoadingProgress</code> Δ	<code>enabled</code> + <code>BusyIndicator</code> Δ	dim style + spinner frame Δ
Input (text)	<code>.ds-input</code>	<code>Input.tsx</code> / <code>Textarea.tsx</code> (files)	<code>OutlinedTextField</code> Δ	<code>TextField</code> + <code>InputDecorator</code> Δ	<code>TextField</code> Δ	<code>TextInput</code> Δ	<code>TextField</code> Δ	<code>textInput</code> bubble Δ (pattern)
Select	<code>.ds-select</code> Δ	<code>Select.tsx</code> (file)	<code>ExposedDropdownMenuBox</code> Δ	<code>DropDownButtonFormField</code> Δ	<code>Picker(menu)</code> Δ	<code>Select</code> Δ	<code>ComboBox</code> Δ	list picker Δ
Checkbox	<code>.ds-check</code> Δ	native + tokens Δ	<code>Checkbox</code> Δ	<code>Checkbox</code> Δ	<code>Toggle.checkbox</code> (macOS) / custom Δ	<code>Checkbox</code> Δ	<code>Checkbox</code> Δ	<code>[x]</code> glyph row Δ
Radio	<code>.ds-radio</code> Δ	native + tokens Δ	<code>RadioButton</code> Δ	<code>Radio</code> Δ	<code>Picker(inline)</code> Δ	<code>Radio</code> Δ	<code>RadioButton</code> Δ	<code>(.)</code> glyph row Δ
Switch		<code>Switch.tsx</code> (file)						

Component	NG Angular .ds-*	RE React	CM Compose	FL Flutter	SW SwiftUI	AR ArkTS	QT Qt/ QML	TU TUI
	<code>.ds-switch</code> o		<code>Switch</code> Δ	<code>Switch.adaptive</code> Δ	<code>Toggle</code> Δ	<code>Toggle(Switch)</code> Δ	<code>Switch</code> Δ	<code>on/off</code> colored token o
Date	<code>.ds-input[type=date]</code> o	native + tokens Δ	<code>DatePickerDialog</code> Δ	<code>showDatePicker</code> Δ	<code>DatePicker</code> Δ	<code>DatePickerDialog</code> Δ	<code>CalendarPopup</code> Δ	<code>YYYY-MM-DD</code> masked input o
Field (label/hint/ error)	<code>.ds-field</code> scaffold o	wrapper + aria Δ	<code>supportingText / isError</code> Δ	<code>InputDecoration(errorText)</code> Δ	<code>.overlay hint text</code> Δ	form hint slot Δ	label + error <code>Text</code> Δ	inline <code>!</code> line o
Card	<code>.ds-card</code> + <code>--raised</code>	<code>Card.tsx</code> (file)	<code>Card / ElevatedCard</code> Δ	<code>Card</code> Δ	<code>GroupBox / custom container</code> Δ	<code>Column</code> + border Δ	<code>Frame / QML rect</code> Δ	bordered box o (pattern)
Stat card (<code>thready-stat</code>)	composite on <code>.ds-card</code> o	composite on <code>Card</code> Δ	composite Δ	composite Δ	composite Δ	composite Δ	composite Δ	big-figure box o
Table (sortable/ paginated)	<code>.ds-table</code> + <code>aria-sort</code> o	table + tokens Δ	<code>LazyColumn</code> + header row Δ	<code>PaginatedDataTable</code> Δ	<code>Table (macOS) / List</code> Δ	<code>List</code> + header Δ	<code>TableView</code> Δ	<code>bubbles/table</code> o
Badge (semantic)	<code>.ds-badge--success/warn/danger</code> (+ <code>--neutral</code> o)	<code>Badge.tsx</code> (file)	tinted <code>Surface</code> chip Δ	<code>Chip / Container</code> tint Δ	<code>Text</code> + capsule bg Δ	<code>Badge</code> Δ	pill <code>Rectangle</code> Δ	glyph <code>✓△×Θ</code> + color o (pattern)
Hashtag chip (direct/ indirect)	<code>thready-tag</code> o	chip on <code>Badge</code> Δ	<code>AssistChip</code> (brand vs outline) Δ	<code>Chip</code> variants Δ	capsule <code>Text</code> variants Δ	<code>Chip</code> Δ	pill + outline Δ	<code>#tag</code> tinted span o
Processing chip (pending/ processing /done/ failed/ retrying)	<code>thready-chip</code> (5 states) o	composite <code>Badge</code> + <code>LoadingSpinner</code> Δ	chip + <code>CircularProgressIndicator</code> Δ	chip + spinner Δ	capsule + <code>ProgressView</code> Δ	chip + progress Δ	pill + <code>BusyIndicator</code> Δ	glyph + spinner frames o

Component	NG Angular <code>.ds-*</code>	RE React	CM Compose	FL Flutter	SW SwiftUI	AR ArkTS	QT Qt/ QML	TU TUI
Topbar nav	<code>.ds-nav</code> + <code>__links/</code> <code>__link</code>	header + tokens Δ	<code>TopAppBar</code> Δ	<code>AppBar</code> Δ	<code>NavigationStack</code> toolbar Δ	<code>Navigation</code> title bar Δ	<code>ToolBar</code> Δ	header line o (pattern)
Sidebar	<code>.ds-</code> <code>sidebar</code> o	nav list Δ	<code>NavigationDrawer</code> / <code>NavigationRail</code> Δ	<code>NavigationRail</code> / <code>Drawer</code> Δ	<code>NavigationSplitView</code> sidebar Δ	<code>SideBarContainer</code> Δ	<code>Drawer</code> Δ	left pane list o
Tabs	<code>.ds-</code> <code>tabs</code> / <code>.ds-</code> <code>tab</code> o	<code>Tabs.tsx</code> (file)	<code>TabRow</code> Δ	<code>TabBar</code> Δ	<code>TabView</code> / <code>Picker</code> (segmented) Δ	<code>Tabs</code> Δ	<code>TabBar</code> Δ	numbered tab line o
Breadcrumbs	<code>.ds-</code> <code>breadcrumbs</code> o	ol + tokens Δ	<code>Row</code> of text buttons Δ	<code>Row</code> of <code>TextButton</code> Δ	<code>HStack</code> links Δ	text row Δ	<code>Row</code> of links Δ	<code>a > b ></code> <code>c</code> line o
Dialog / modal	<code>.ds-dialog</code> on <code><dialog></code> o	portal + tokens Δ	<code>AlertDialog</code> Δ	<code>showDialog</code> / <code>AlertDialog</code> Δ	<code>.sheet</code> / <code>.customDialog</code> Δ	<code>Dialog</code> Δ	<code>Dialog</code> Δ	centered overlay box o
Toast / alert	<code>thready-</code> <code>toast</code> + <code>.ds-alert</code> o	region + tokens Δ	<code>SnackBar</code> / <code>SnackBarHost</code> Δ	<code>SnackBar</code> Δ	custom overlay / <code>.action</code> Δ	<code>prompt</code> <code>.showToast</code> Δ	overlay <code>Popup</code> Δ	<code>notifications.go</code> pane (pattern)
Progress bar (det./ indet./ failed+retry)	<code>.ds-</code> <code>progress</code> o	<code>Progress.tsx</code> (file)	<code>LinearProgressIndicator</code> Δ	<code>LinearProgressIndicator</code> Δ	<code>ProgressView</code> (value:) Δ	<code>Progress</code> Δ	<code>ProgressBar</code> Δ	 bar + <code>[r]etry</code> o (pattern)
Spinner (ring)	<code>.ds-</code> <code>spinner</code> o	<code>LoadingSpinner.tsx</code> (file)	<code>CircularProgressIndicator</code> Δ	<code>CircularProgressIndicator</code> Δ	<code>ProgressView</code> (Δ	<code>LoadingProgress</code> Δ	<code>BusyIndicator</code> Δ	<code>bubbles/</code> <code>spinner</code> frames o
Helix-motif spinner	inline SVG (2 strands + rungs) o	same SVG component Δ	<code>Canvas</code> bezier strands Δ	<code>CustomPainter</code> Δ	<code>Canvas</code> / <code>Path</code> shape Δ	<code>Canvas</code> drawing Δ	QML <code>Canvas</code> / <code>Shape</code> Δ	braille/ ASCII helix frames <code>...</code> o
Avatar (sizes/	<code>.ds-avatar</code> o	<code>Avatar.tsx</code> (file)					rounded	

Component	NG Angular .ds-*	RE React	CM Compose	FL Flutter	SW SwiftUI	AR ArkTS	QT Qt/ QML	TU TUI
status/ group)			<code>Box</code> circle + initials Δ	<code>CircleAvatar</code> Δ	<code>Circle</code> + initials overlay Δ	<code>Image /</code> circle text Δ	<code>Rectangle</code> Δ	(MV) initials token ○
Empty state	<code>thready- empty</code> ○	<code>EmptyState.tsx</code> (file)	centered <code>Column</code> composite Δ	centered <code>Column</code> Δ	<code>ContentUnavailableView</code> Δ	empty composite Δ	centered composite Δ	centered text + action key ○
Error state (page + retry)	<code>thready- error</code> on <code>thready- empty</code> ○	<code>ErrorBoundary.tsx</code> (file)	error composite Δ	error composite Δ	error view Δ	error composite Δ	error composite Δ	danger text + <code>[r]etry</code> ○
Skeleton loader	<code>.ds- skeleton</code> ○	shimmer div Δ	Modifier shimmer / placeholder Δ	shimmer container Δ	<code>.redacted(reason: .placeholder)</code> Δ	opacity -pulse rows Δ	animated rects Δ	... placeholder (no shimmer) ○
Pagination	<code>.ds- pagination</code> ○	button row Δ	button <code>Row</code> Δ	button <code>Row</code> Δ	<code>HStack</code> buttons Δ	button row Δ	<code>Row</code> of buttons Δ	< 1 [2] 3 > line ○
Tooltip	<code>.ds-tip</code> (hover+focus) ○	title/popover + tokens Δ	<code>PlainText</code> Δ	<code>Tooltip</code> widget Δ	<code>.help()</code> (macOS) / popover Δ	<code>Popup</code> hint Δ	<code>Tooltip</code> Δ	status-bar hint line ○
Link	<code>.ds-link</code>	anchor + tokens Δ	<code>ClickableText</code> Δ	<code>TextButton</code> -styled text Δ	<code>Link</code> Δ	<code>Span</code> + click Δ	<code>Text</code> + <code>MouseEvent</code> Δ	underlined accent text ○
Footer (attribution)	<code>.ds-footer</code> + <code>FooterComponent</code>	footer + tokens Δ	bottom bar text Δ	bottom text Δ	bottom <code>Text</code> Δ	bottom text Δ	status text Δ	bottom line ○ (pattern)
Theme toggle	<code>ds-theme- toggle</code> (Angular cmp)	toggle + <code>data-theme</code> stamp Δ	<code>isSystemInDarkTheme()</code> + pref Δ	<code>ThemeMode</code> setting Δ	<code>preferredColorScheme</code> Δ	dark- mode config Δ	palette swap Δ	terminal palette (dark default) ○
Language picker	<code>ds- language- picker</code>	select + i18n dict Δ	locale dropdown Δ	<code>Locale</code> selector Δ	<code>Picker</code> + <code>Locale</code> Δ	locale select Δ	<code>ComboBox</code> Δ	<code>lang:</code> selector ○ (i18n.go pattern)

Component	NG Angular ds-*	RE React	CM Compose	FL Flutter	SW SwiftUI	AR ArkTS	QT Qt/ QML	TU TUI
	(Angular cmp)							

Reading rule. A in RE means the **file exists at HEAD** — it does *not* override the `SCAFFOLD/FLAGGED` status of the whole package `[GAP: 8.6]`; re-audit before use. NG cells are shipped, production-usable CSS/Angular code. Every CM/FL/SW/AR/QT cell is ASSUMED because the packages are verified scaffolds or unverified: they name the intended native widget so the token-bridge codegen (design-system.md §7) has a target, nothing more.

4. Per-component platform customization notes

- **Buttons.** Web keeps the verified `:focus-visible` `--focus-ring`; Compose/Flutter map it to their own focus indication (`Modifier.focusable` ripple, `FocusNode` highlight) rather than a literal 3px ring. TUI: primary = accent background + `--accent-on` ink (`lipgloss.Style.Bold`), ghost = accent foreground only; disabled = `--muted` fg. Destructive stays `--danger` on **every** platform — brand color must never mask an error `[VERIFIED rule — THEMES.md]`.
- **Inputs.** Native pickers win on mobile: date uses the platform sheet (Compose `DatePickerDialog`, SwiftUI `DatePicker`, Flutter `showDatePicker`) — the web `input[type=date]` look is *not* replicated. Error semantics travel as a triple (border/danger tint, message text, `aria-invalid` / `isError` / `errorText`) so the AA meter and form validation read identically.
- **Select / radio / switch.** iOS convention prefers `Picker` over radio groups — the SwiftUI cell for “radio” is an inline picker by design (documented divergence, same semantics). ArkTS `Toggle(ToggleType.Switch)` and Qt `Switch` take the accent track color from the token bridge.
- **Table.** Sorting is a header-owned state (`aria-sort` on web, `sortAscending` in Flutter’s `PaginatedDataTable`, column click in Qt `TableView`). On phones the table collapses to thread-row cards (wireframes pattern) — Compose/Flutter/SwiftUI ship the card list, not a squeezed table. TUI uses `bubbles/table` with Lipgloss header style.
- **Hashtag chip.** The direct/indirect distinction must never be color-only: web appends the visually-hidden “(derived)”, Compose/Flutter/SwiftUI append it to the semantics/ `Semantics` label, TUI renders indirect tags dimmed **and** suffixed `~`. Direct chips use `--brand` fill + `--brand-ink`; indirect use outline + `--muted`.
- **Processing chips / pipeline.** The five states (pending/processing/done/failed/retrying) and the precedence caption are a cross-platform parity contract (component-library.md §7): every platform exposes a keyboard-focusable (or key-bound, TUI `[r]`) idempotent retry on `failed`.
- **Dialogs.** Mobile prefers native alert/sheet; Esc/back-gesture dismissal maps per platform. The destructive action keeps `--danger` and is never the default-focused control.

- **Toasts.** Web uses `role=status/alert` live regions; Compose `SnackbarHost`, Flutter `ScaffoldMessenger`, ArkTS `promptAction.showToast`, TUI the verified `notifications.go` pane. A toast is never the only channel for a critical error (also inline) — all platforms.
- **Helix-motif spinner.** Vector strands (`--accent` + `--brand-2`) rotate on capable platforms; the TUI substitutes a frame-cycled braille helix; `prefers-reduced-motion` (and its platform equivalents: `UIAccessibility.isReduceMotionEnabled`, Android `ANIMATOR_DURATION_SCALE`) freezes it to a static glyph.
- **Skeletons.** Shimmer is web/Compose/Flutter; SwiftUI uses `.redacted`; TUI uses static `...` placeholders (a terminal shimmer is noise). All appear only after ~150 ms (no flash).
- **Tooltips.** Touch platforms have no hover: Compose/Flutter/ArkTS show tooltips on long-press, SwiftUI on pointer hover (iPadOS) — hints that matter must also exist as visible helper text.
- **Theme & brand axes.** Every platform consumes the same two axes (theming.md): mode (light/dark/system) via the platform's dark-mode API, brand via the generated token binding (Lipgloss palette, `ThreadyColors`, etc.) fetched from `/v1/accounts/{id}/branding`.

5. Verified vs. assumed summary

Column	VERIFIED cells	PROPOSED cells	ASSUMED cells	Basis
Angular <code>.ds-*</code>	10 of 36 component rows ride verbatim shipped CSS/ Angular (<code>container/section</code> , <code>btn</code> ×3 variants, <code>card</code> , <code>input</code> , <code>link</code> , <code>nav</code> , <code>footer</code> , <code>badge</code> ×3, <code>brand-mark</code> , <code>theme-toggle</code> , <code>language-picker</code>)	26 rows are PROPOSED extensions authored in <code>components.html</code>	0	file content fetched
React	13 component files verified at HEAD (<code>Avatar</code> , <code>Badge</code> , <code>Button</code> , <code>Card</code> , <code>EmptyState</code> , <code>ErrorBoundary</code> , <code>Input</code> , <code>LoadingSpinner</code> , <code>Progress</code> , <code>Select</code> , <code>Switch</code> , <code>Tabs</code> , <code>Textarea</code>) — file existence only, package FLAGGED [GAP: 8.6]	—	all remaining rows (table, dialog, toast, tooltip, pagination, checkbox, radio, date, breadcrumbs, sidebar, chips...)	tree fetched
Compose (KMP)	<code>Theme.kt</code> / <code>Animations.kt</code> / <code>Accessibility.kt</code> utilities exist (theme bridge target)	—	all widget rows — no components exist [GAP: 8.4]	tree fetched
Flutter (helix_design)	repo exists but is an empty scaffold (verified)	—	all rows [GAP: 8.2/8.3]	tree fetched
SwiftUI	—	—	all rows — no in-house package located [OPEN: THREADY-DES-LIB-02]	search
ArkTS	—	—	all rows (native via <code>helix_shims</code> , uninspected) [GAP: 8.5]	ground truth
Qt/QML	—	—	all rows (native via <code>helix_shims</code> , uninspected) [GAP: 8.5]	ground truth
TUI (Lipgloss)	Bubble Tea + Lipgloss pattern verified in local	Thready style set (generated from	—	local clone read

Column	VERIFIED cells	PROPOSED cells	ASSUMED cells	Basis
	<code>helix_track</code> clone (incl. <code>notifications.go</code> , <code>i18n.go</code>)	token export) — all rows PROPOSED on the verified pattern		

Bottom line (matches the ground-truth honesty rule): only the **Angular web layer** and the **TUI pattern** are grounded in verified, usable in-house code today. React has verified files but a flagged package. Compose, Flutter, SwiftUI, ArkTS and Qt columns are intent, not capability.

6. Open items

- `[OPEN: THREADY-DES-11]` (inherited) — final public API/class names for all PROPOSED `.ds-*` extensions and `thready-*` composites must be agreed with the web team before upstreaming.
- `[OPEN: THREADY-DES-12]` (inherited) — decide which PROPOSED components (select, switch, table, dialog, toast, progress, spinner, avatar, skeleton, pagination, tooltip, tabs, breadcrumbs, sidebar) are generic enough to contribute upstream to `design_system` vs. keep in Thready.
- `[OPEN: THREADY-DES-LIB-01]` — the React column must be re-audited component-by-component (`[GAP: 8.6]`): confirm each verified file actually implements the states in this library (hover/focus/disabled/loading/error) before any React surface consumes it.
- `[OPEN: THREADY-DES-LIB-02]` — no in-house SwiftUI package exists or is named anywhere in the decision matrix; decide whether iOS ships via KMP/Compose (current plan) making the SwiftUI column moot, or a SwiftUI package becomes a `[BUILD-NEW]` item.
- `[OPEN: THREADY-DES-LIB-03]` — the `helix_shims` interface for ArkTS/Qt was not inspected (`[GAP: 8.5]`); the AR/QT cells need re-verification once the shim contracts exist.
- `[OPEN: THREADY-DES-LIB-04]` — the token-bridge codegen (CSS → Lipgloss / `ThreadyColors` / Flutter theme) referenced by every non-web cell does not exist yet; tracked as THREADY-DES-KMP-01 / -FLUT-01 / -TUI-01 in `component-library.md` §10.

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